Histology of skin: Epidermis, Dermis, & Hypodermis

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Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

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1. Identify the general divisions of the skin.
2. Distinguish the different types of regional variations of the skin.
3. Describe the development, organization, and functions of keratinocytes.
4. Identify the other cells of the epidermis, their spatial arrangement, and functions.
5. Contrast the two dermal layers along with their associated structures

Relevant previous lectures:
FOM, Introduction to Histology Part II –Tissues, Fibers, Cells
FOM, Histology of the Epithelium
PPOM1, General Sensory Systems

Skin (aka Integument) is a complex organ

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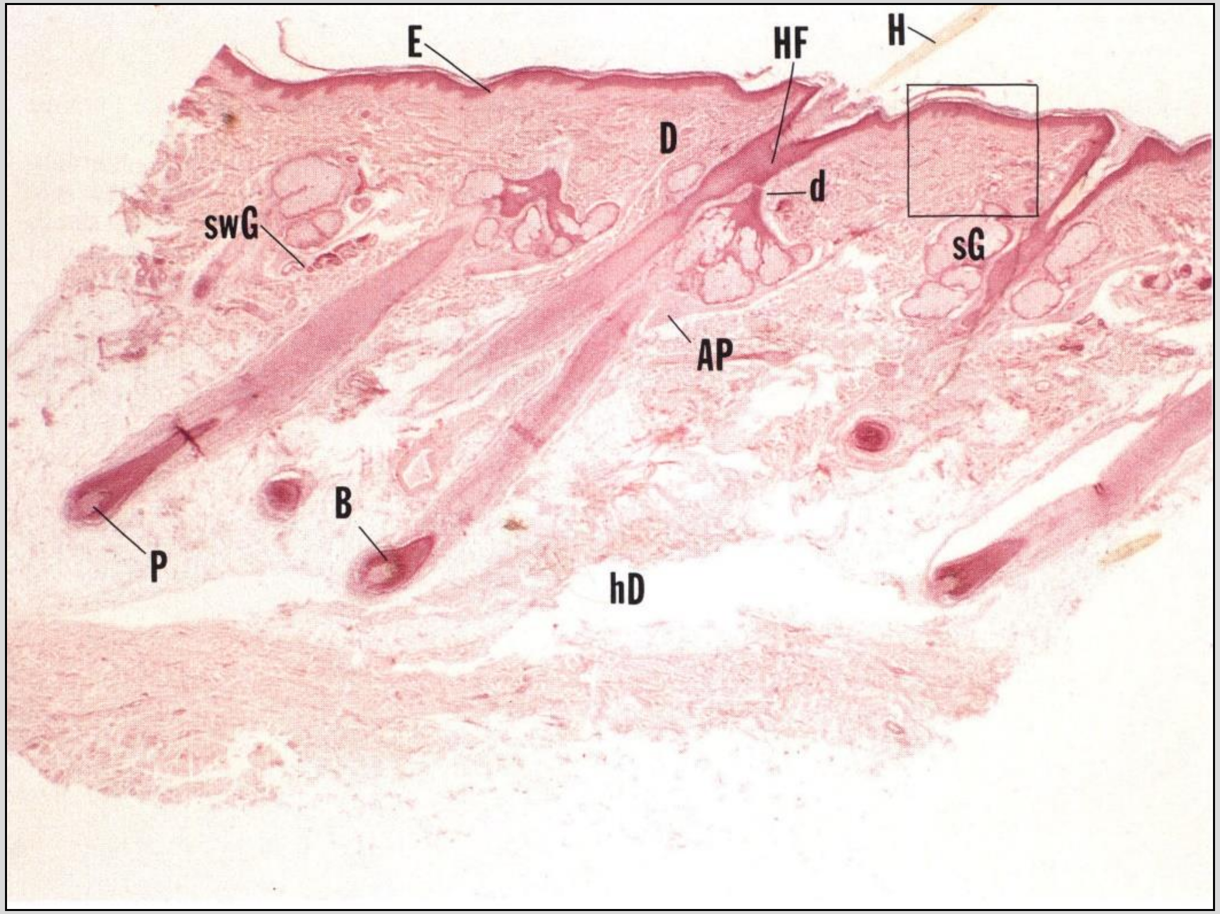
Multi-layered covering

Regional specialization

Wide variety of functions:

- <u>Barrier</u>: H_20, mechanical, & UV light
- <u>Immunological</u>: bacterial
- <u>Endocrine</u>: Vitamin D
- <u>Exocrine</u>: secretion of sweat & sebum,
- <u>Homeostasis</u>: temperature regulation
- <u>Sensory</u>: touch, heat/cold, pain



Source: "Integument" in LP Gartner's Textbook and Atlas of Histology, 7th Ed. (2018)

General Divisions of Skin

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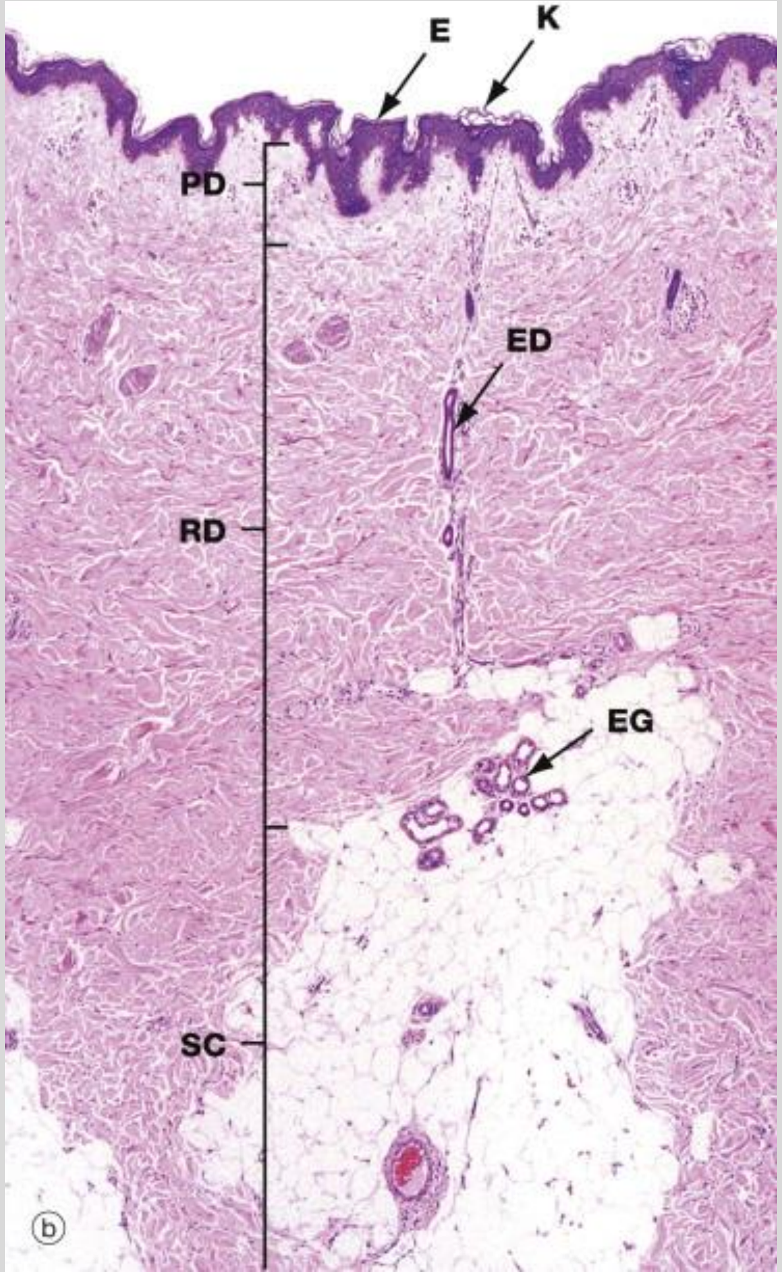


Epidermis

- keratinized stratified squamous epithelium
- derived from ectoderm
- avascular
- attached to basement membrane

Dermis

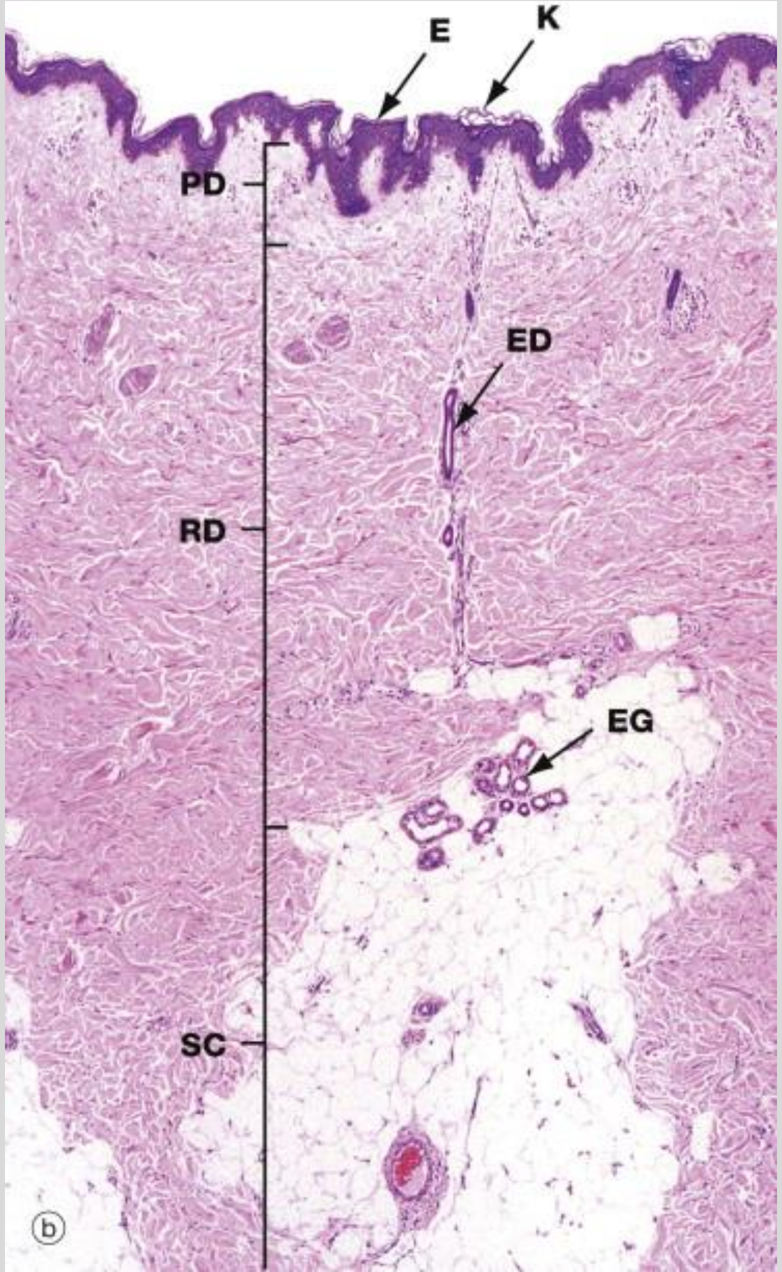
- dense connective tissue
- derived from mesoderm
- Highly vascular

Hypodermis

- subcutaneous fascia composed of septa (fibrous sheets) and adipose
- derived from mesoderm
- vascular

Source: “Skin” in Wheater’s Functional Histology, 6th Ed. (2014)

General Divisions of Skin



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<u>Epidermal Appendages</u>

- Found throughout the layers of the skin
- Include:
  - Hair
  - Sweat glands
  - Sebaceous glands
  - Nails
  - Mammary glands

Source: “Skin” in Wheater’s Functional Histology, 6th Ed. (2014)

Histological definition: Thick versus Thin skin

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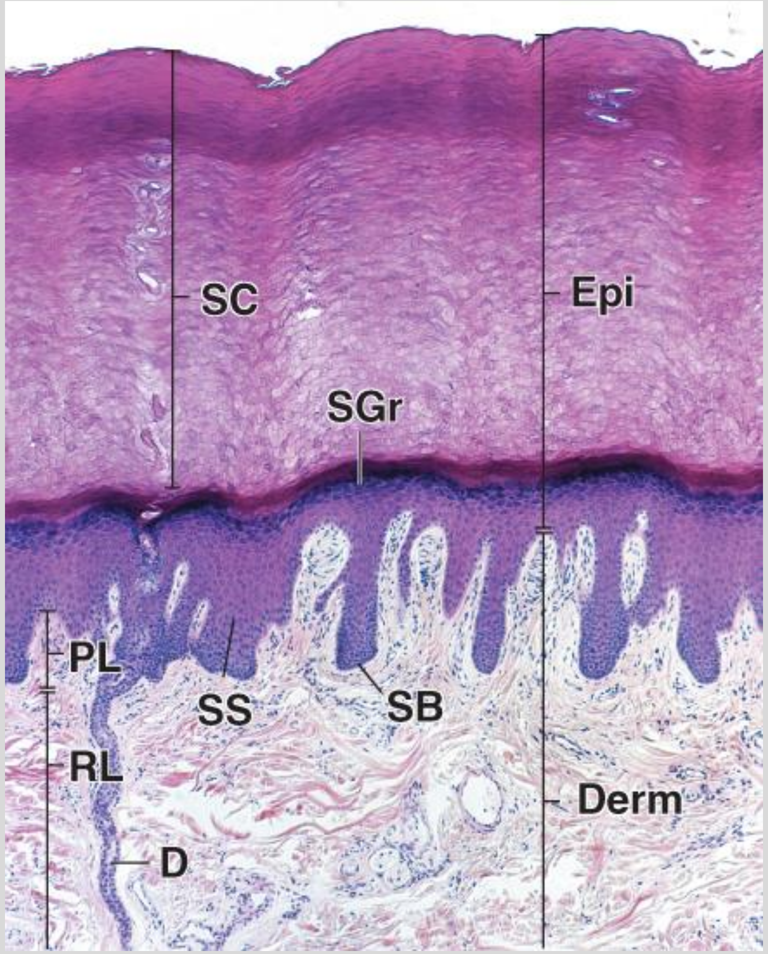

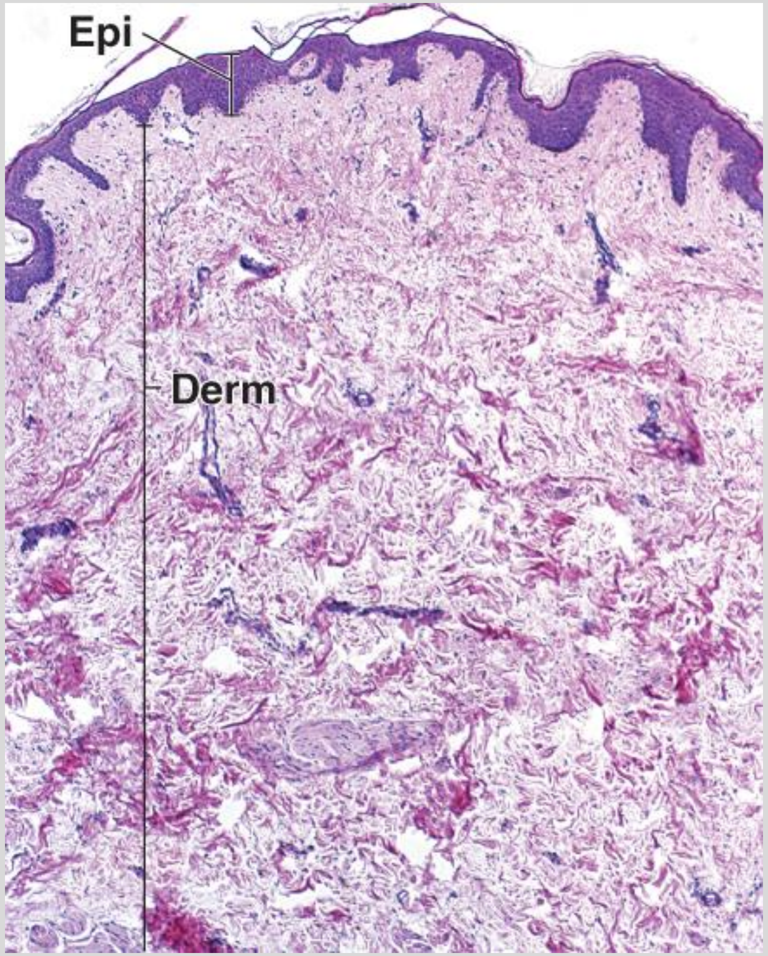

Source: “Integumentary System” in Ross and Pawlina Histology, 7th Ed. (2015)

<u>Thick Skin</u>

- epidermis is very thick
- Skin of palms and soles
    - These areas have increased mechanical interaction with the world, so the thick epidermis acts as a mechanical barrier

<u>Thin Skin</u>

- epidermis is thin
- Everywhere else
    - Yes, that includes both the eyelids and the back!

Histological definition: Glabrous versus Non-Glabrous skin

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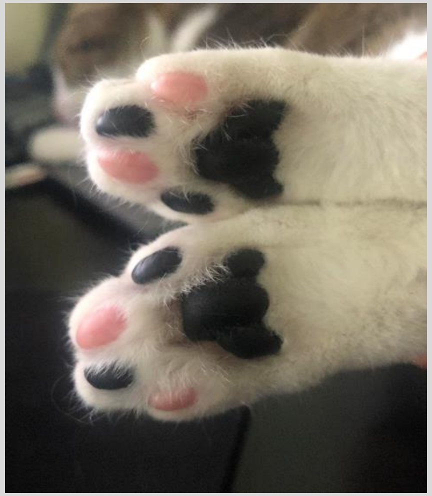
Glabrous – hairless skin

Non-glabrous – hairy skin

human palm

cat pads



Putting it together:

Thick skin ←→ Glabrous

Thin skin ←→ Non-Glabrous

Source: "Integumentary System" in Ross and Pawlina Histology, 7th Ed. (2015)

Keratinocytes

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- Predominate cell type of epidermis (85%+)
- Originate in deepest layer of epidermis and grow outward, then get sloughed off
- Produce keratin, a structural protein of epithelial cells in all vertebrates
- Formation of epidermal water barrier.

Skin is continuously developed and shed/desquamated

Versus

Layers of the Epidermis

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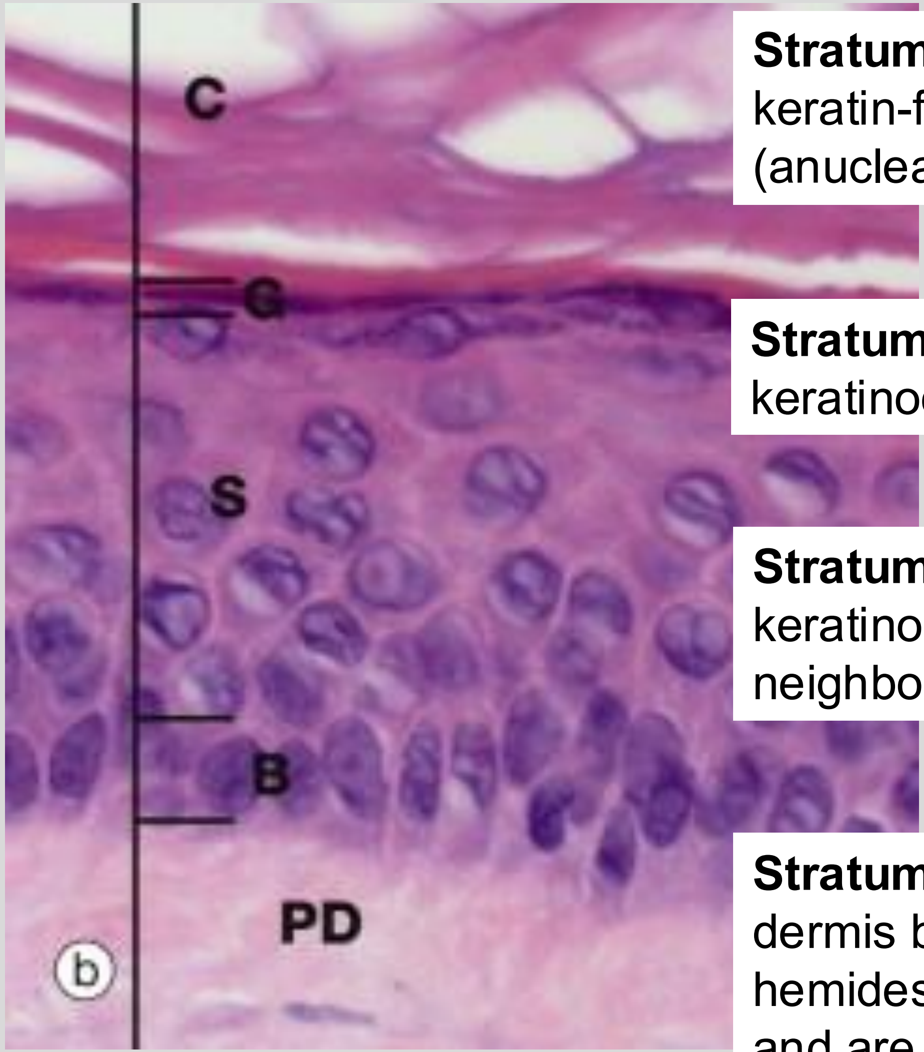


Stratum corneum – outermost layer, made up of keratin-filled dead keratinocytes termed squames (anucleate); layer which varies most in thickness.

Stratum lucidum – thin layer of cells between SC and SGr, only evident in thick skin, considered subdivision of SC.

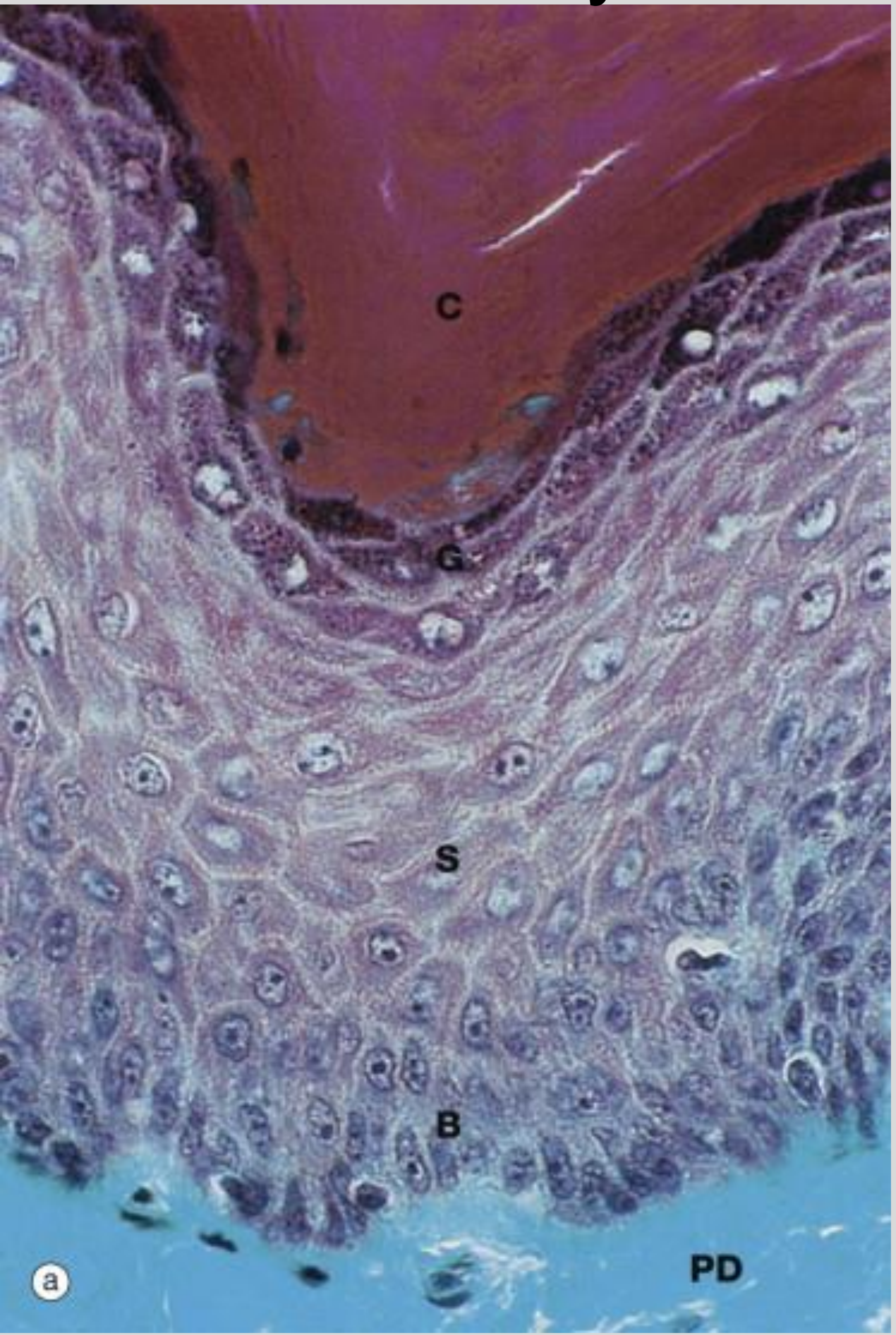
Stratum granulosum – diamond shaped keratinocytes, contain keratohyalin granules

Stratum spinosum – irregular, polyhedral keratinocytes with desmosomes that link neighboring cells (look like “spines”).

Stratum basale – deepest layer, separated from dermis by basement membrane and attached by hemidesmosomes. Cells are cuboidal to columnar and are mostly (mitotically active) keratinocytes.

Source: “Skin” in Wheater’s Functional Histology, 6th Ed. (2014)

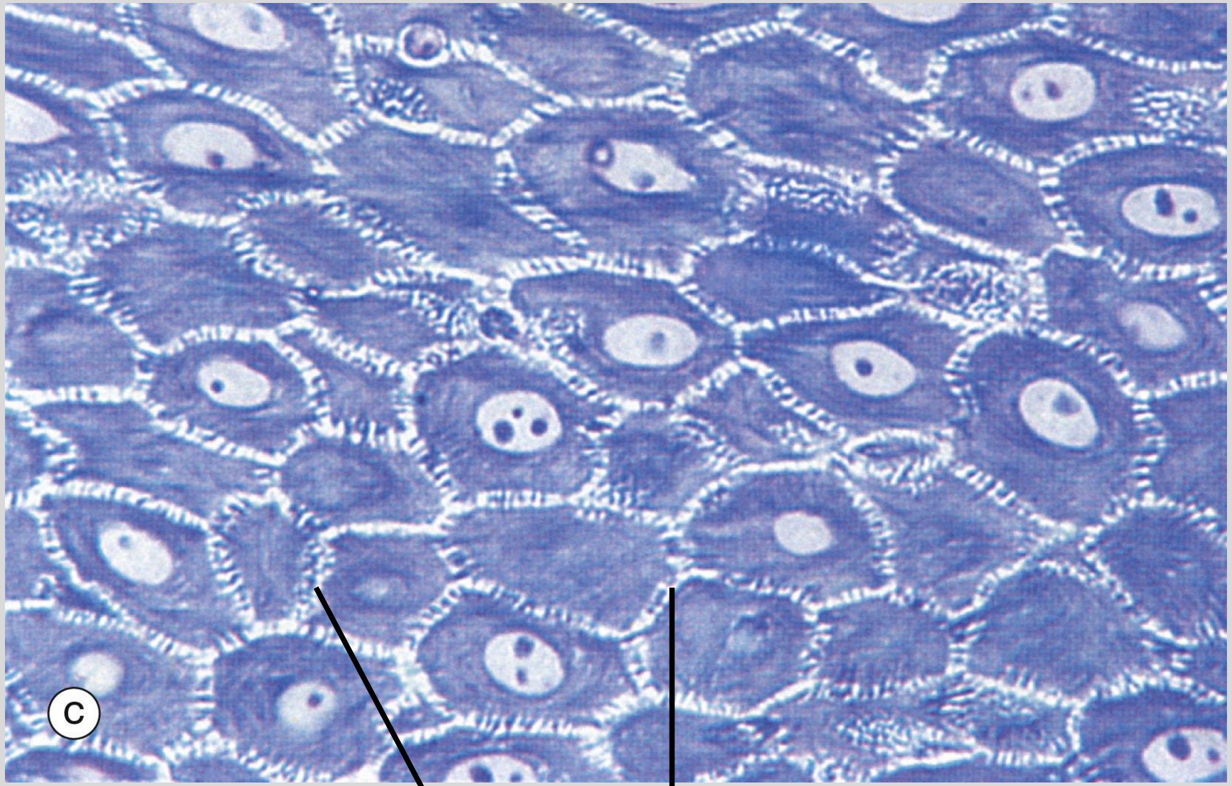
Layers of the Epidermis



C stratum corneum
G stratum granulosum

S stratum spinosum

B stratum basale



desmosomes

Source: "Skin" in Wheater's Functional Histology, 7th Ed. (2024)

Development and migration of keratinocytes

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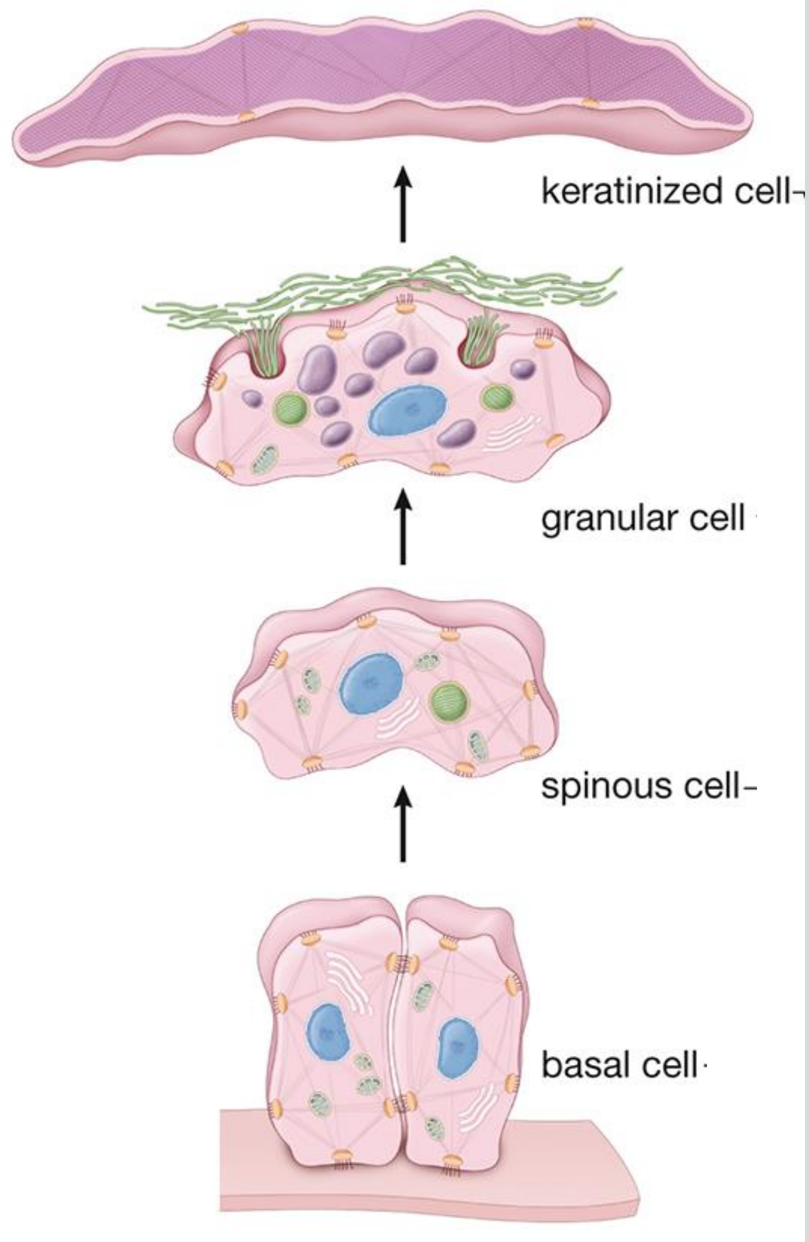

Cells enter stratum corneum, lose organelles (incl. lamellar bodies) and nuclei. The keratohyalin granules turn tonofibrils into a homogenous keratin matrix. No metabolic activity

Cells pushed into stratum granulosum and become flattened. The cells accumulate keratohyalin granules mixed between tonofibrils.

KCs are pushed into stratum spinosum. Begin producing keratohyalin granules (contain filaggrin and trichohyalin) to aggregate keratin filaments and conversion to cornified cells. KCs also produce lamellar bodies.

Basal KC begins synthesis of keratin (organized into groups called Tonofilaments) which are grouped into bundles (Tonofibrils). The tonofibrils are inserted into desmosomes

Source: "Integumentary System" in Ross and Pawlina Histology, 7th Ed. (2024)

Development and migration of keratinocytes

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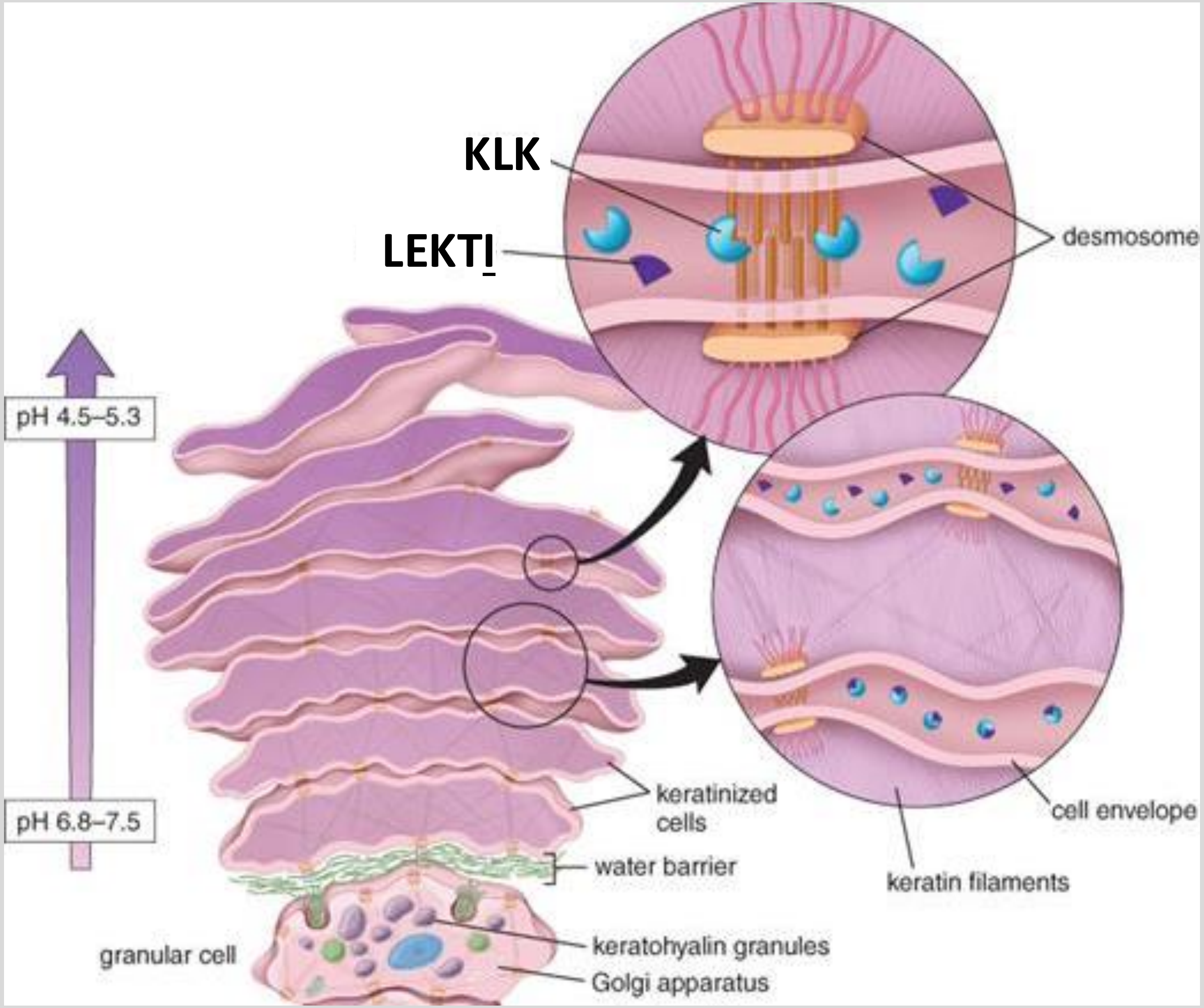

Cornified cells are desquamated via break-down of desmosomes. Proteinase activity of a resident enzyme of the interstitial fluid (KLK, kallikrein-related serine peptidase) is triggered by lowered pH near the surface.

Lymphoepithelial Kazal-type Inhibitor Inhibits KLK. It dissociates when the pH decreases

Source: "Integumentary System" in Ross and Pawlina Histology, 7th Ed. (2015)

Keratinization Timeline

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Birth of daughter cell in S. Basale

↓

Stratum Spinosum

↓

Stratum Granulosum

30 Days

↓ 2 to 6 hours

Stratum Corneum (fully keratinized)

↓ 15 Days

Desquamated

Psoriasis/Epidermal hyperplasia

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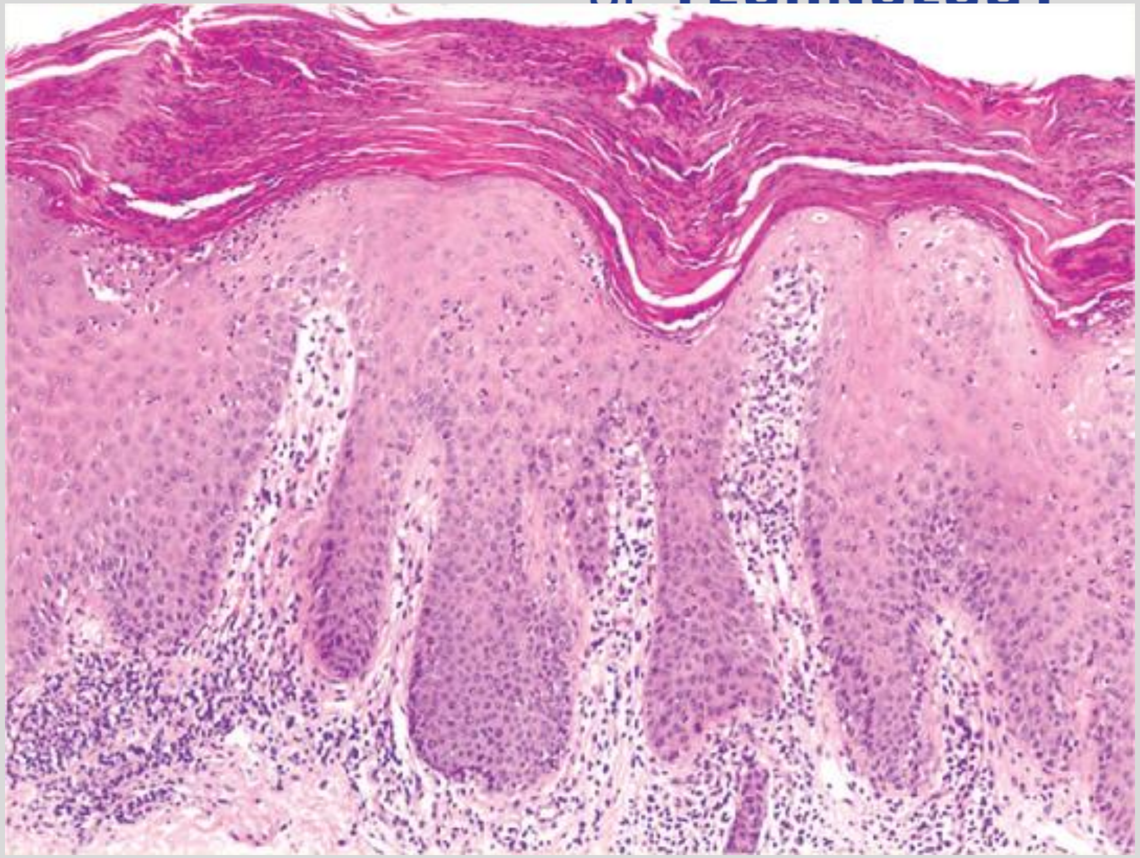
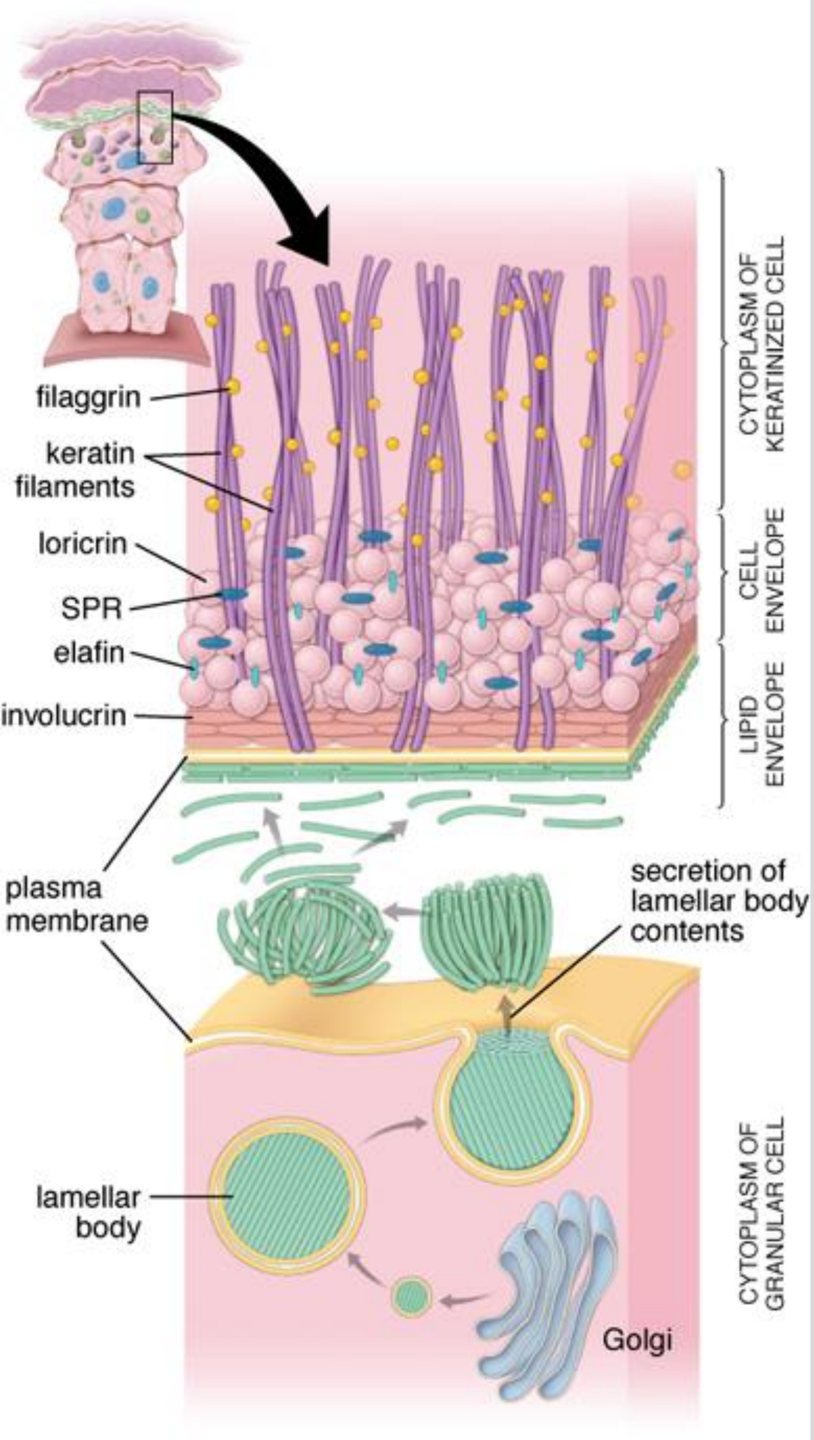
Source: “Boehncke and Schön (2015) Psoriasis The Lancet



Gartner and Hiatt's Atlas and Text of Histology, 8e, 2023

Accelerated maturation of keratinocytes, 8-10 days.
Insufficient development of tonofibrils and keratohyaline.
Granular layer does not form normally.
Skin surface has white flakes over thick red epidermis.



Epidermal water barrier

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Cell envelope

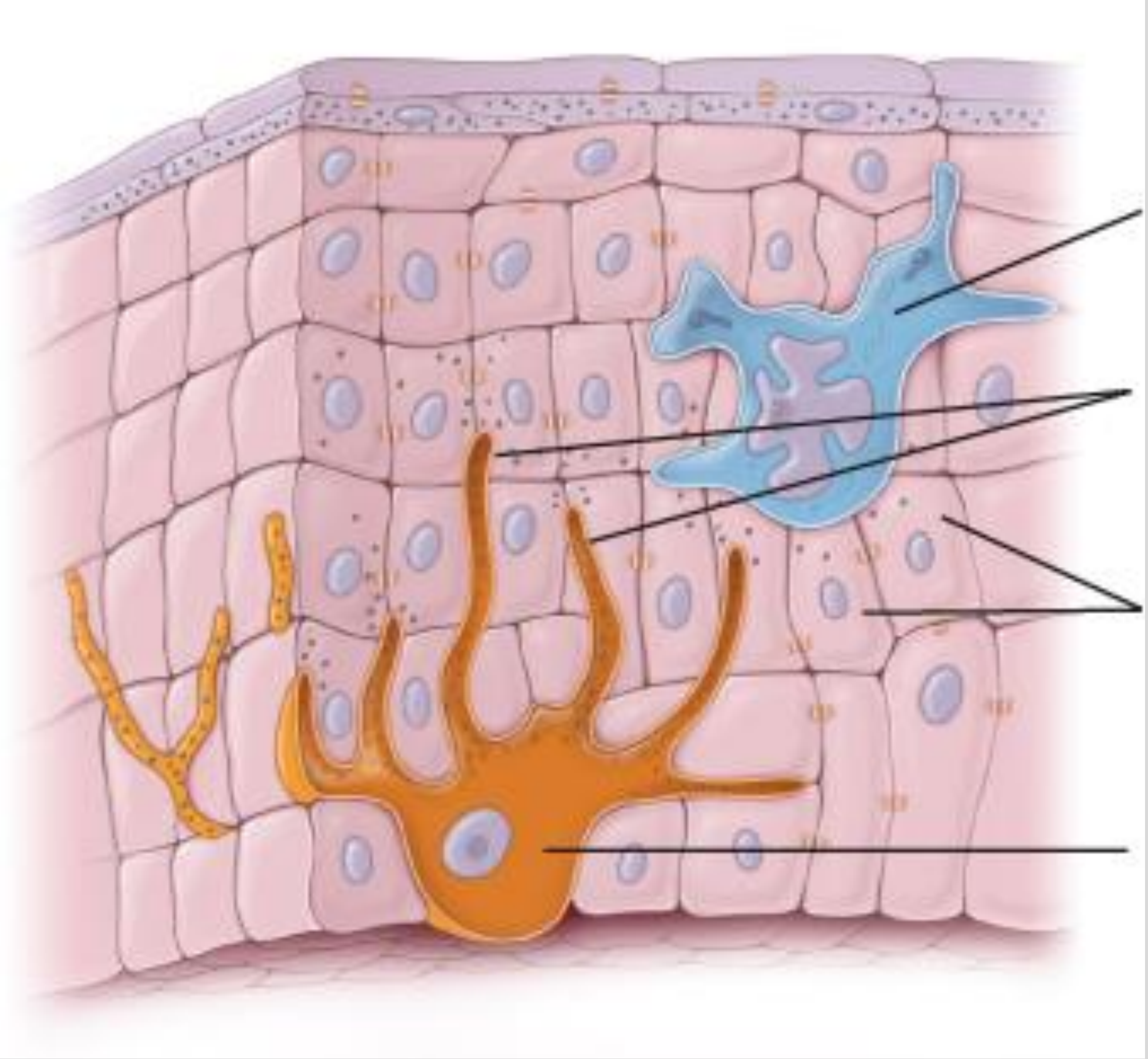
- Lattice of insoluble proteins on *inner surface* of plasma membrane.
- Cross-linking of small proline-rich proteins and larger proteins esp. loricrin (85% of all of the proteins)
- Contributes to strong mechanics of barrier.

Lipid envelope

- Lipid/hydrophobic layer attached to *outer surface* of plasma membrane.
- Lamellar bodies (produced in Stratum Spinosum) contain mixture of oils.
  - Exocytosed between the stratum granulosum and corneum and release the oils.

Source: “Integumentary System” in Ross and Pawlina Histology, 7th Ed. (2015)

Other cells of the epidermis



Langherhan's cells (immune cells)

Keratinocytes (structural cells)

Melanocytes (pigment-synthesizing cells)

*also Merkel's cells (sensory cells), next lecture

Source: "Integumentary System" in Ross and Pawlina Histology, 7th Ed. (2015)

Melanocytes

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- Located between the cells of the Stratum Basale
- Derived from Neural Crest cells
- Make pigment (melanin) in melanosomes
- Provide melanin granules to keratinocytes through a process known as “pigment donation”
- Melanin protects against Ultraviolet (UV) radiation

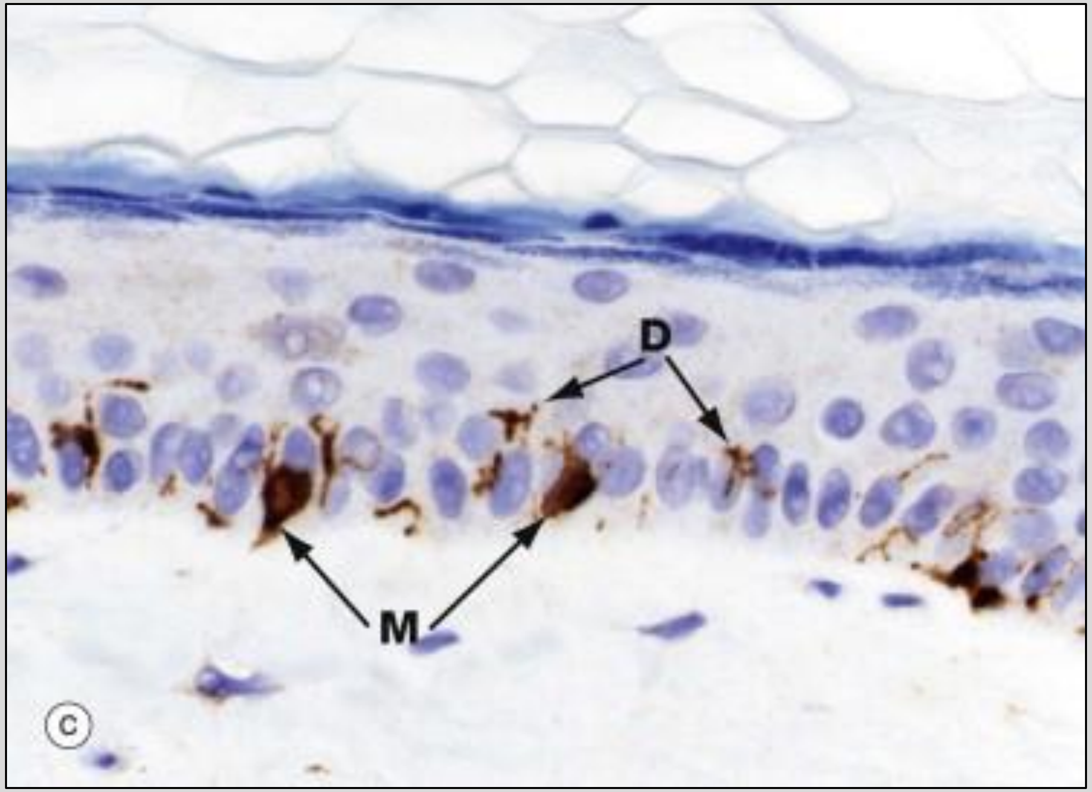
Tyrosine

↓

L-DOPA

↓

Melanin



Source: “Skin” in Wheater’s Functional Histology, 6th Ed. (2014)
“Integumentary System” in Ross and Pawlina Histology, 9th Ed. (2024)

Pigment Donation

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Daughter cell that moves up the skin layers phagocytoses a portion of the melanocyte (as seen in circles 4,5,6 in the image)

↓

The phagocytosed melanosome continues to make melanin in the keratinocyte's cytoplasm

↓

The melanocyte's membrane breaks down

↓

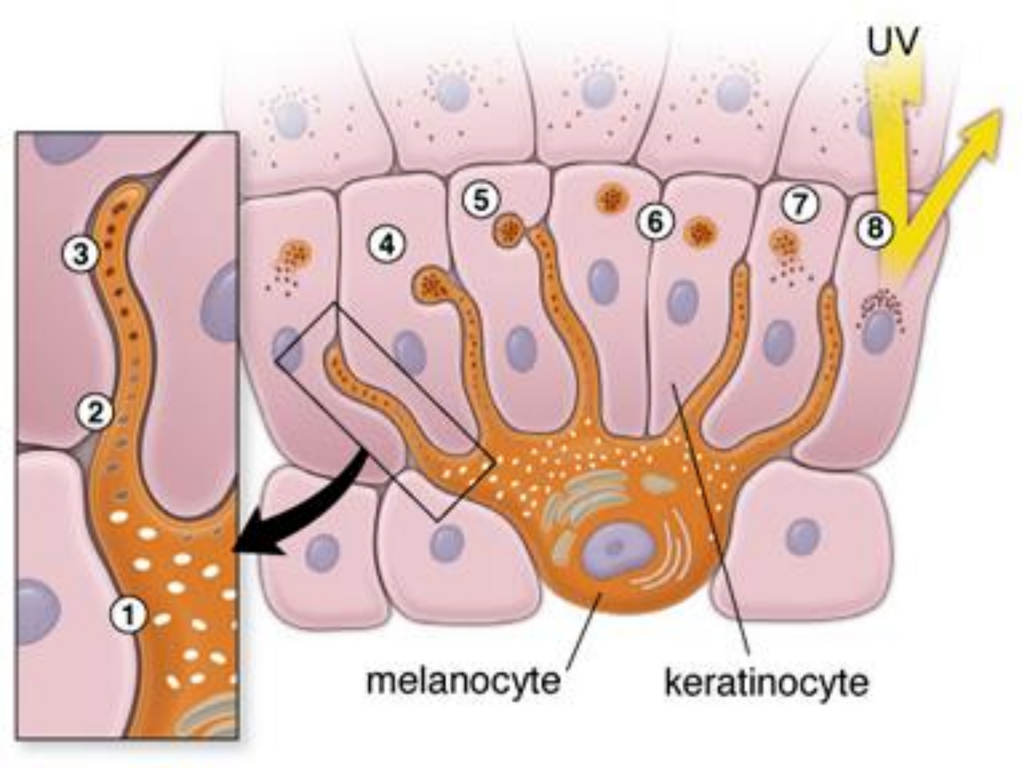
The pigment (melanin) is released into the keratinocyte's cytoplasm

↓

The melanin covers the nucleus to form a "cap"

↓

The nucleus is protected



Source: "Skin" in Wheater's Functional Histology, 6th Ed. (2014)
"Integumentary System" in Ross and Pawlina Histology, 9th Ed. (2024)

Two disorders involving melanocytes

<u>Albinism</u>
- Genetic disorder (autosomal recessive)
- Evident in neonates
- Commonly involves eyes and skin
- Lack of mechanism to develop melanin
    - Have melanocytes, but they don't make melanin

<u>Vitiligo</u>
- Autoimmune disorder
- Lack of melanocytes
- Affects select portions of the skin
- expressed at any time in life
- Usually otherwise healthy
    - 20% comorbid with other autoimmune diseases
        - Especially thyroid
- The immune system attacks melanocytes
    - → melanocyte death

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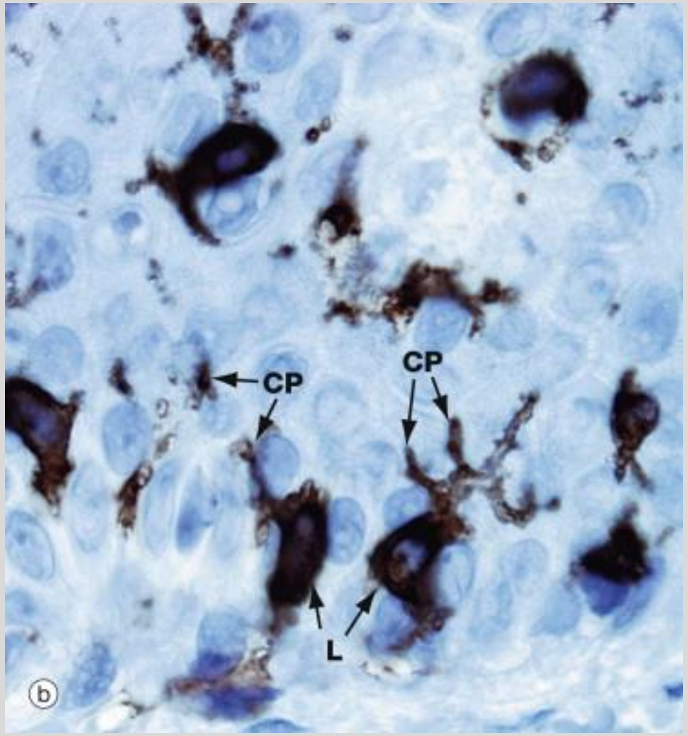
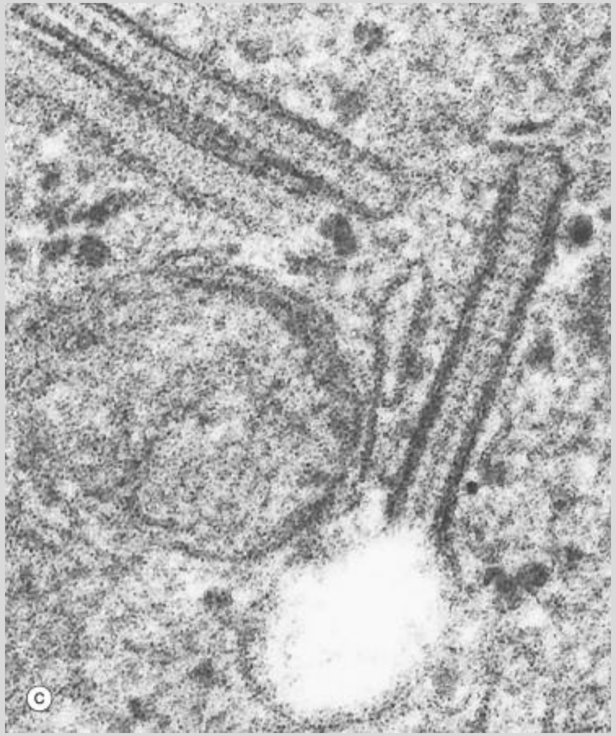
<u>Source</u>:

https://www.bbc.com/news/world-asia-china-56464881

https://my.clevelandclinic.org/health/diseases/12419-vitiligo

Langerhans's cells





- Immune system cells (Antigen-presenting cells)
- Located mostly in the Stratum Spinosum
- Mesenchymal origin
    - derived from bone marrow stem cells
    - part of the mononuclear-phagocytic system
- Have Cytoplasmic processes (CP)
    - tendrils that reach out, retract, and move in different directions
        - sample the space around them
        - look for antigens
    - express MHC1 and MHC2 molecules
- Mobile

<u>Birbeck Granules</u>

- Organelle only found in the Langerhans's Cell
- Function: enclose and degrade pathogens

Source: “Skin” in Wheater’s Functional Histology, 6th Ed. (2014)

What happens when a Langerhans's cells finds an antigen?

Find an antigen

↓

Dissolve the Basement Membrane via enzymes

↓

Leave the Epidermis

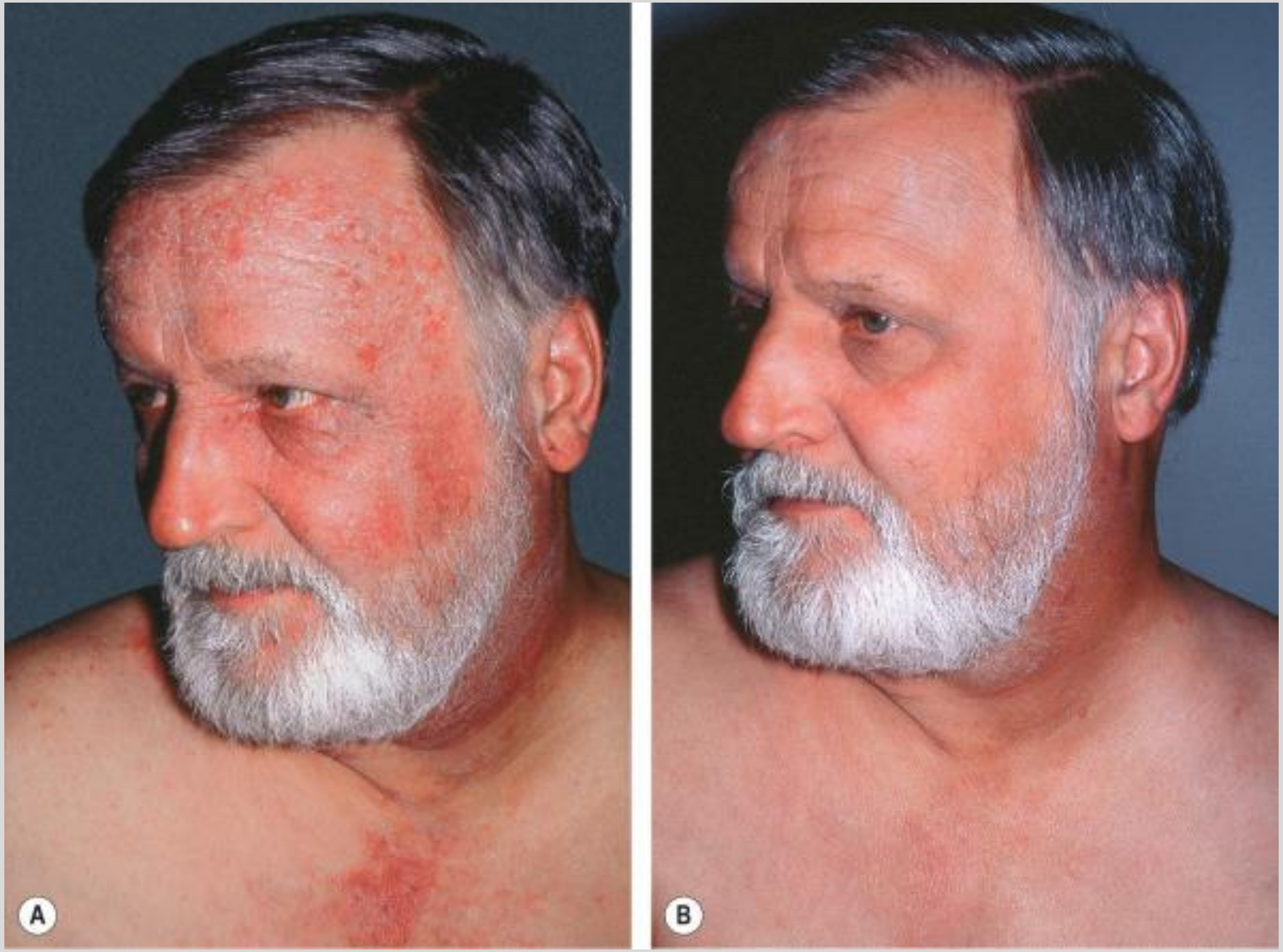
↓

Enter the lymphatic system

↓

Present the antigen to a T-Cell

Example of Langerhans's cells – What they can do!



Involved in delayed-type hypersensitivity reactions
e.g. contact allergic dermatitis

Dermal-epidermal junction

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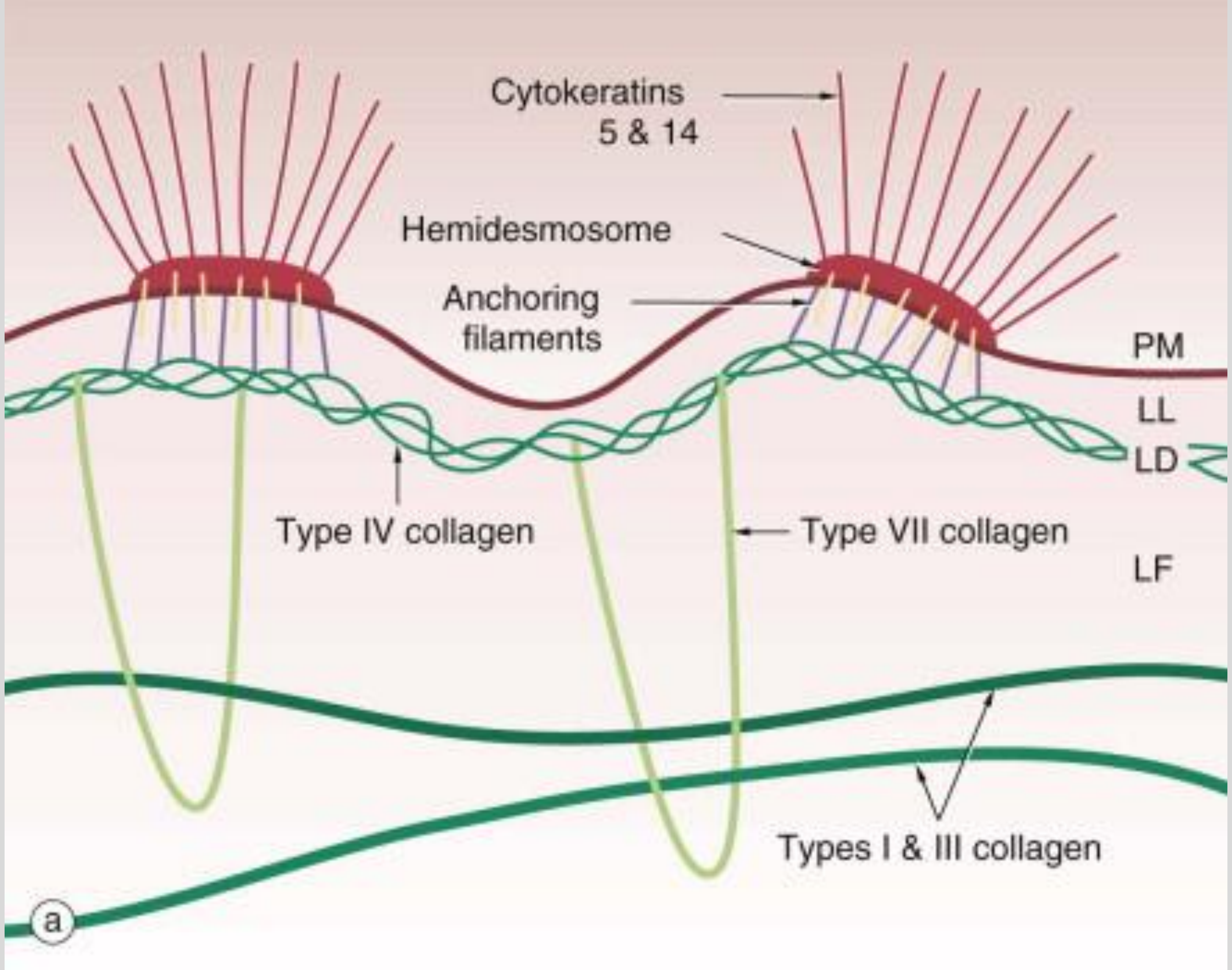

*HD = Hemidesmosome

Source: “Skin” in Wheater’s Functional Histology, 7th Ed. (2024)

The basement membrane is at the junction of the epidermis, like all epithelial tissue.

Anchoring filaments attach HD* to collagen fibers, which link the HD to papillary dermis.

Dermal-epidermal junction

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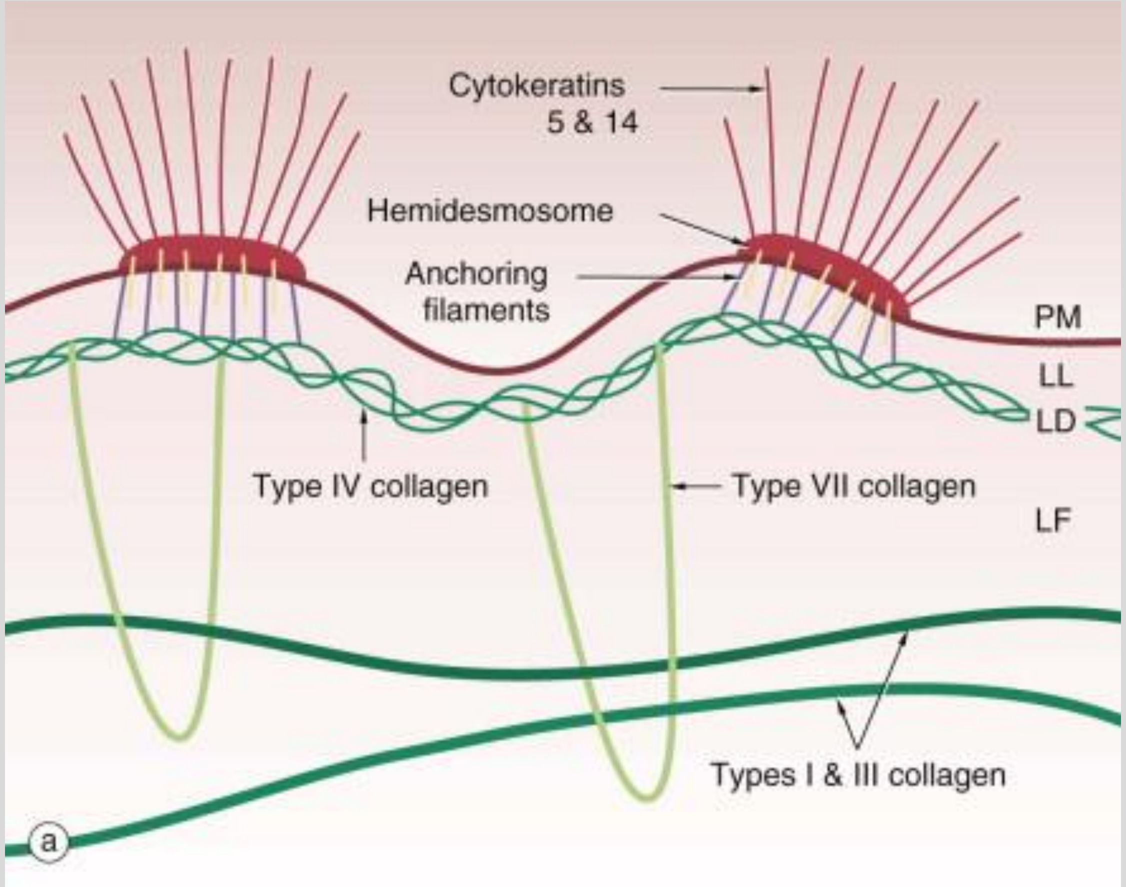
The epidermis is connected to the dermis via hemidesmosomes

The basement membrane is at the junction of the epidermis, like all epithelial tissue. It is composed of collagen.

The cytokeratin seen in the picture is made of tonofilaments

- These tonofilaments attach to the hemidesmosomes in the keratinocyte's plasma membrane.
- Anchoring filaments connect to the Lamina Densa in the Basement Membrane

The Basement Membrane is attached to the collagen fibers of the papillary dermis via loops of Type 7 collagen



Source: "Skin" in Wheater's Functional Histology, 7th Ed. (2024)

SEM of the Dermal-epidermal junction

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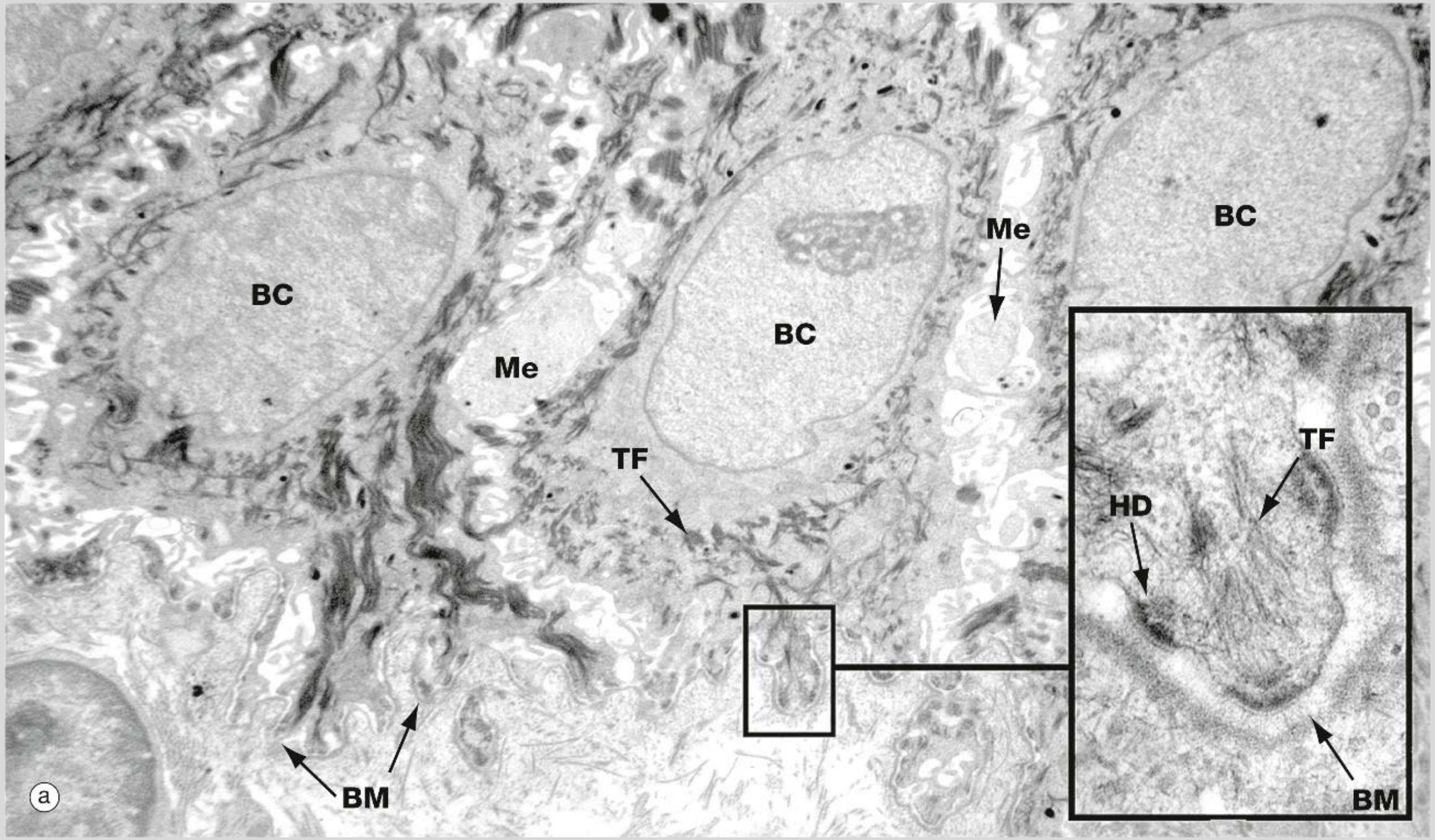

Source: “Skin” in Wheater’s Functional Histology, 7th Ed. (2024)

BC = Basal Cell

Me = Melanocyte

BM = Basement Membrane

TF = Tonofilament

HD = Hemidesmosome

Dermis

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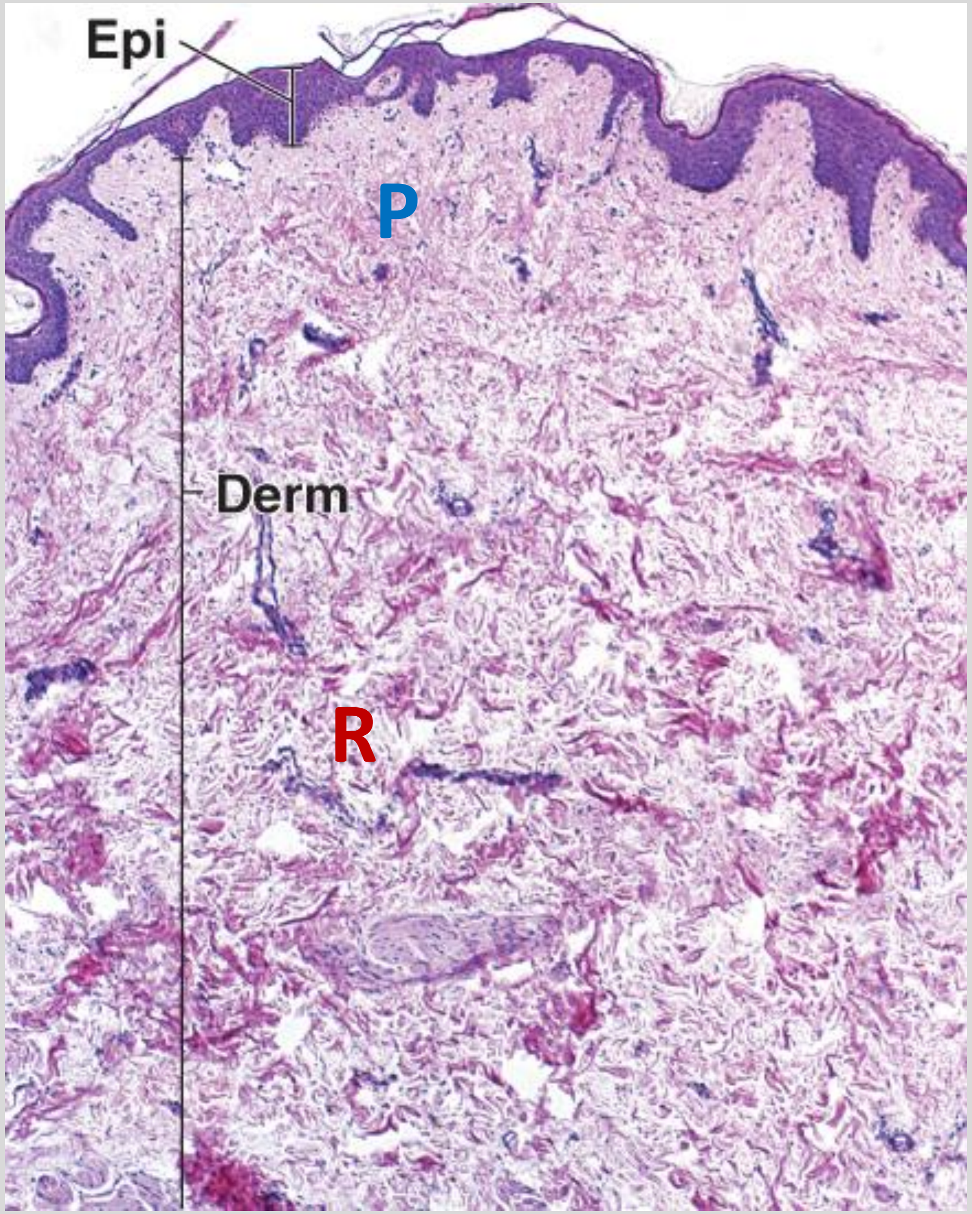

Connective tissue derived from mesoderm

Vascularized

Houses the epidermal appendages and many sensory neurons

2 Divisions

- Due to and correspond to the types of connective tissue found in the layers
- No dividing layer between them

Source: "Integumentary System" in Ross and Pawlina Histology, 7th Ed. (2015)

Two layers of the dermis

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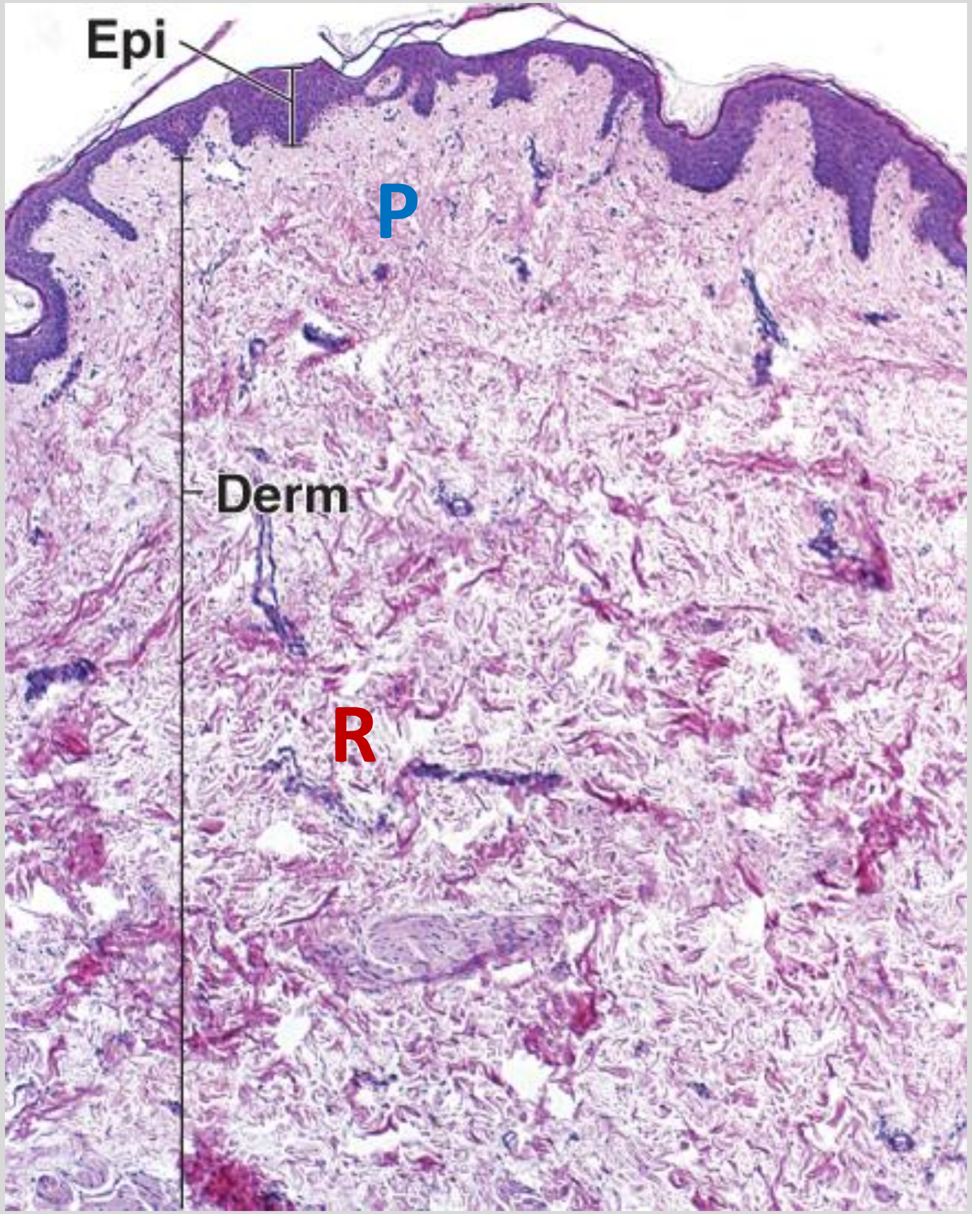


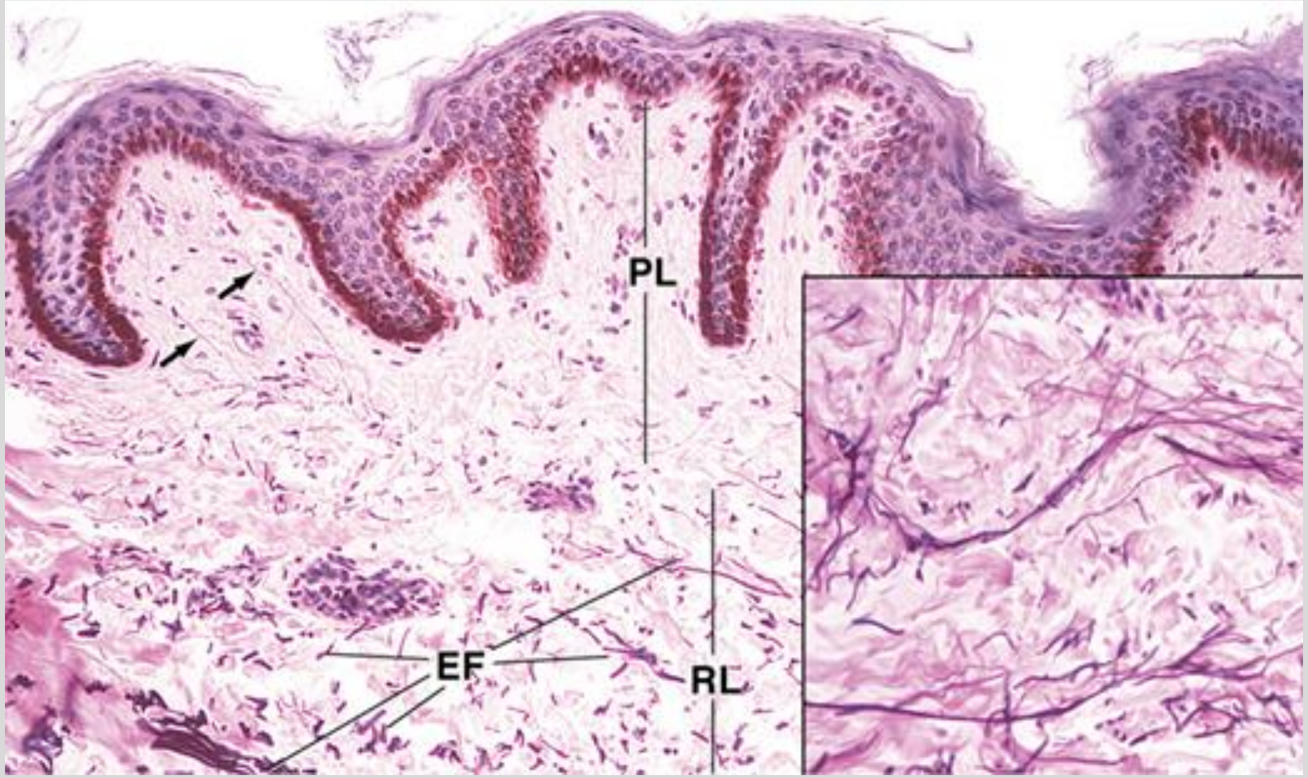
Papillary Layer

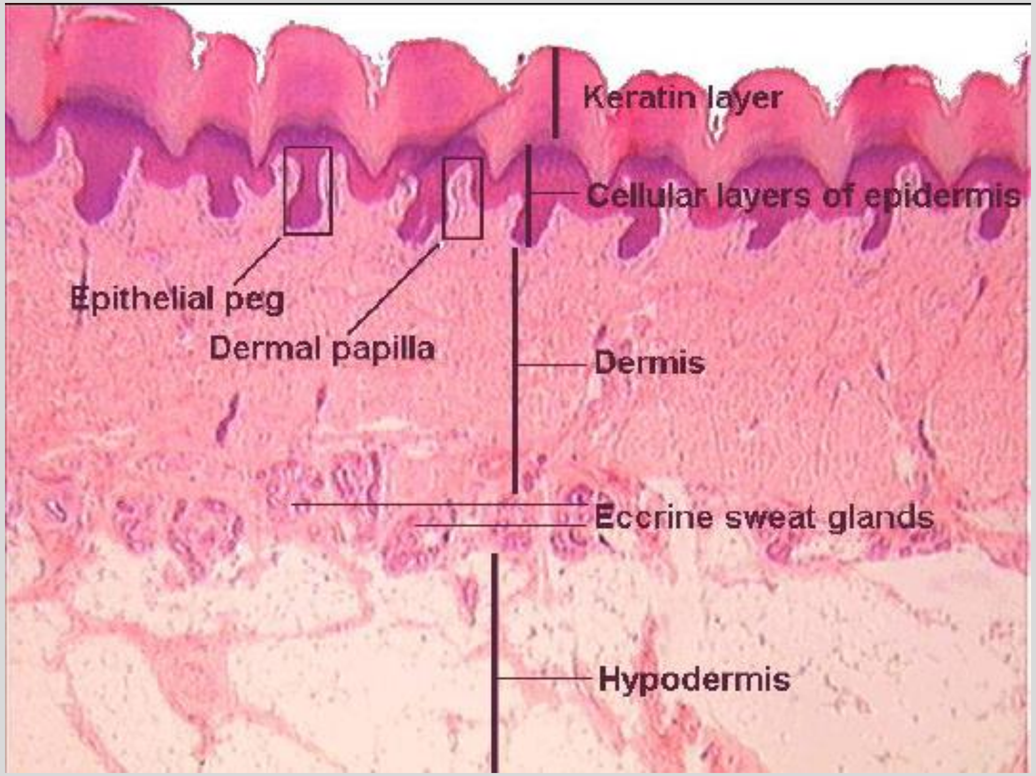
- Outer layer connected to the basement membrane
- Composed of loose connective tissue (areolar)
- Lots of ground substance, Type 1 and Type 3 collagen
- Thinner, paler

Reticular Layer

- Deeper layer
- Less cellular
- Composed of dense elastic and irregular connective tissue
- Will also find bundles of Type 1 collagen fibers here
- Thicker, darker, coarser, larger

Two layers of the dermis



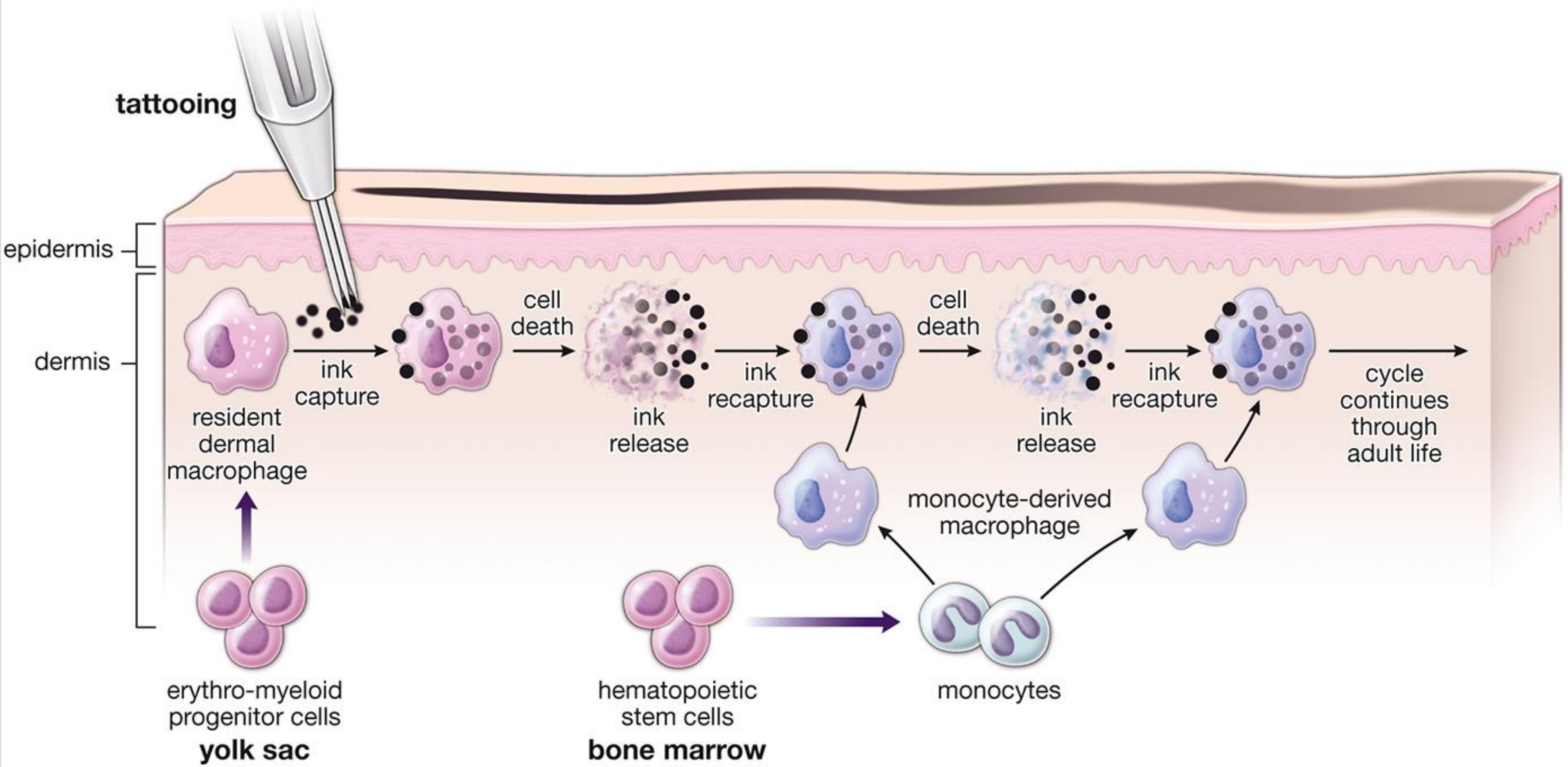


Note: Epidermal Peg = "Rete Ridge"

Tight physical linkage of dermis & epidermis

Source: "Integumentary System" in Ross and Pawlina Histology, 6th Ed. (2010)

Tattoos



Ink is injected into the Papillary Dermis layer. It is phagocytized by a macrophage and kept until the cell's death. Cycle repeats. Blurry over time by re-distribution of ink across successive generations of cells. Can be cleared if ink particles are pulverized (esp. laser removal).

Source: "Integumentary System" in Ross and Pawlina Histology, 9th Ed. (2024)

Dermis is highly vascularized

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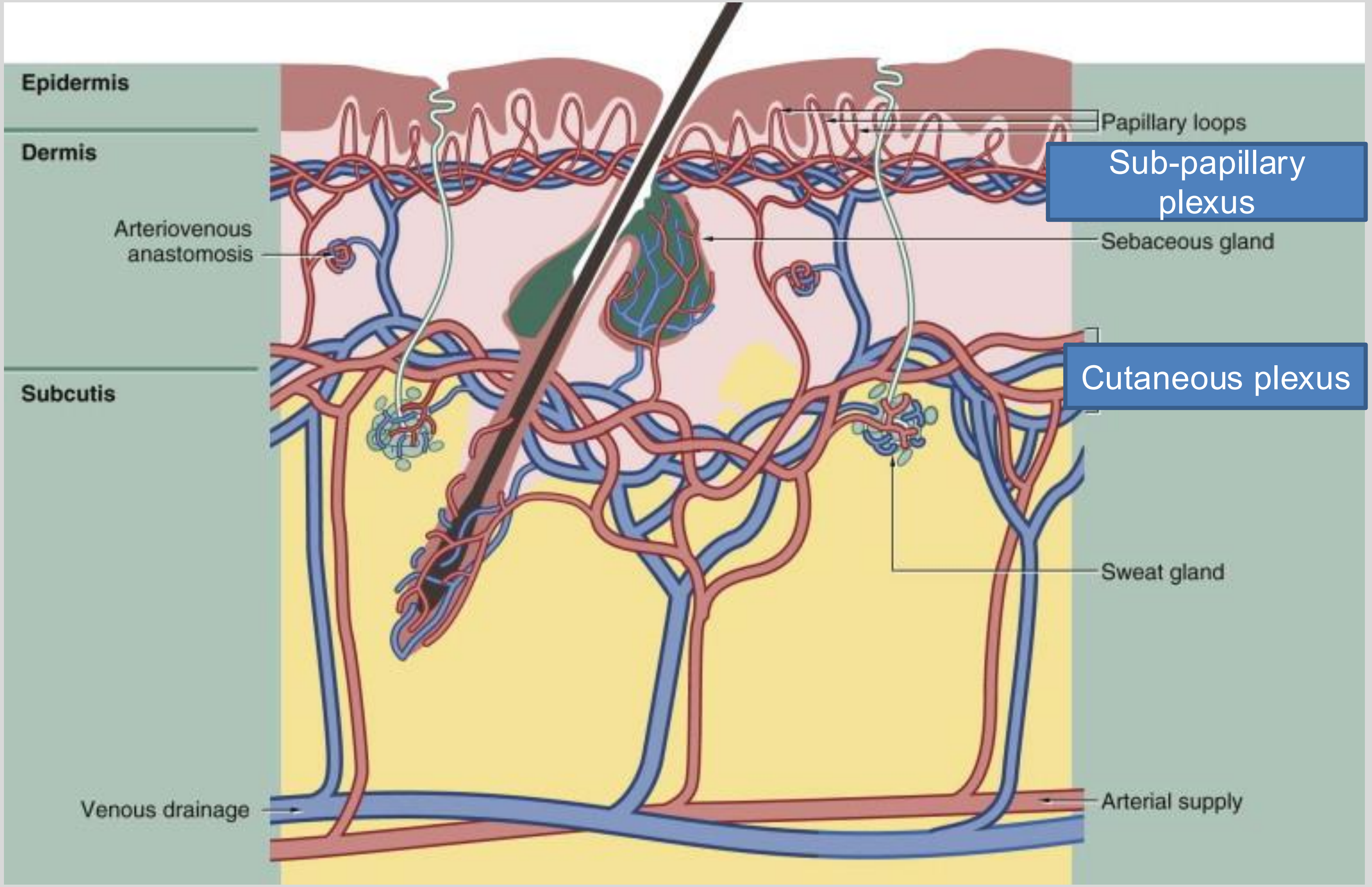

<u>Sub-papillary Plexus</u>

- Most peripheral
- Has loops that enter each dermal papilla to supply the above dermis

<u>Cutaneous Plexus</u>

- Just above the hypodermis
- In the lowest layer of the Reticular Dermis
- Supplies the various epidermal appendages

Source: “Skin” in Wheater’s Functional Histology, 6th Ed. (2014)

Dermis is highly vascularized

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Note the circular structures. These are vessels that surround a gland or hair follicle

Source: “Skin and its appendages” in Gray’s Anatomy (2016)

Dermis also includes lymphatics

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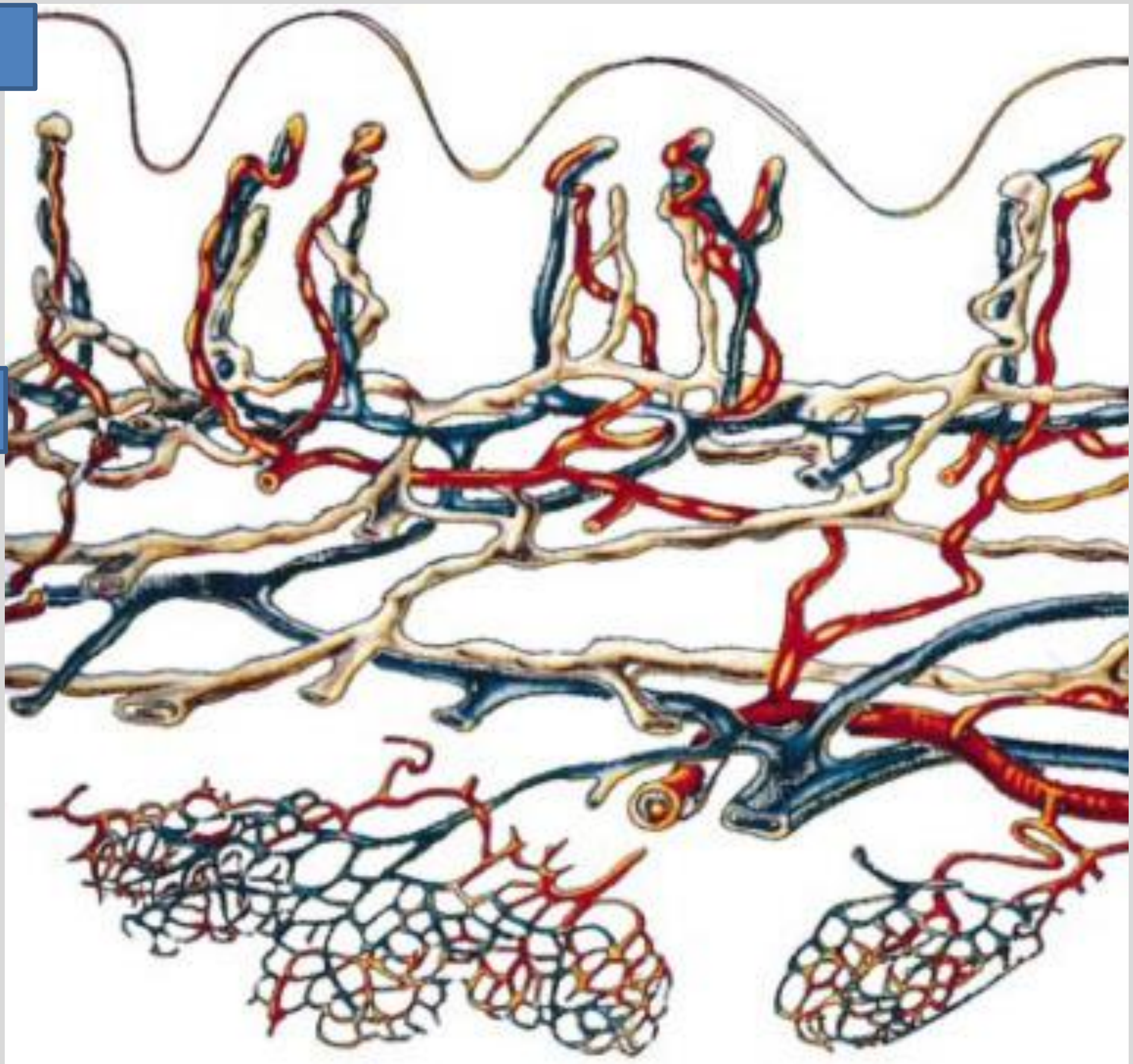
Dermal Papilla

Sub-papillary plexus



The lymphatic system runs parallel with the vasculature

Source: Journal of Investigative Dermatology Symposium Proceedings (2000)

Langer Lines of reticular dermis

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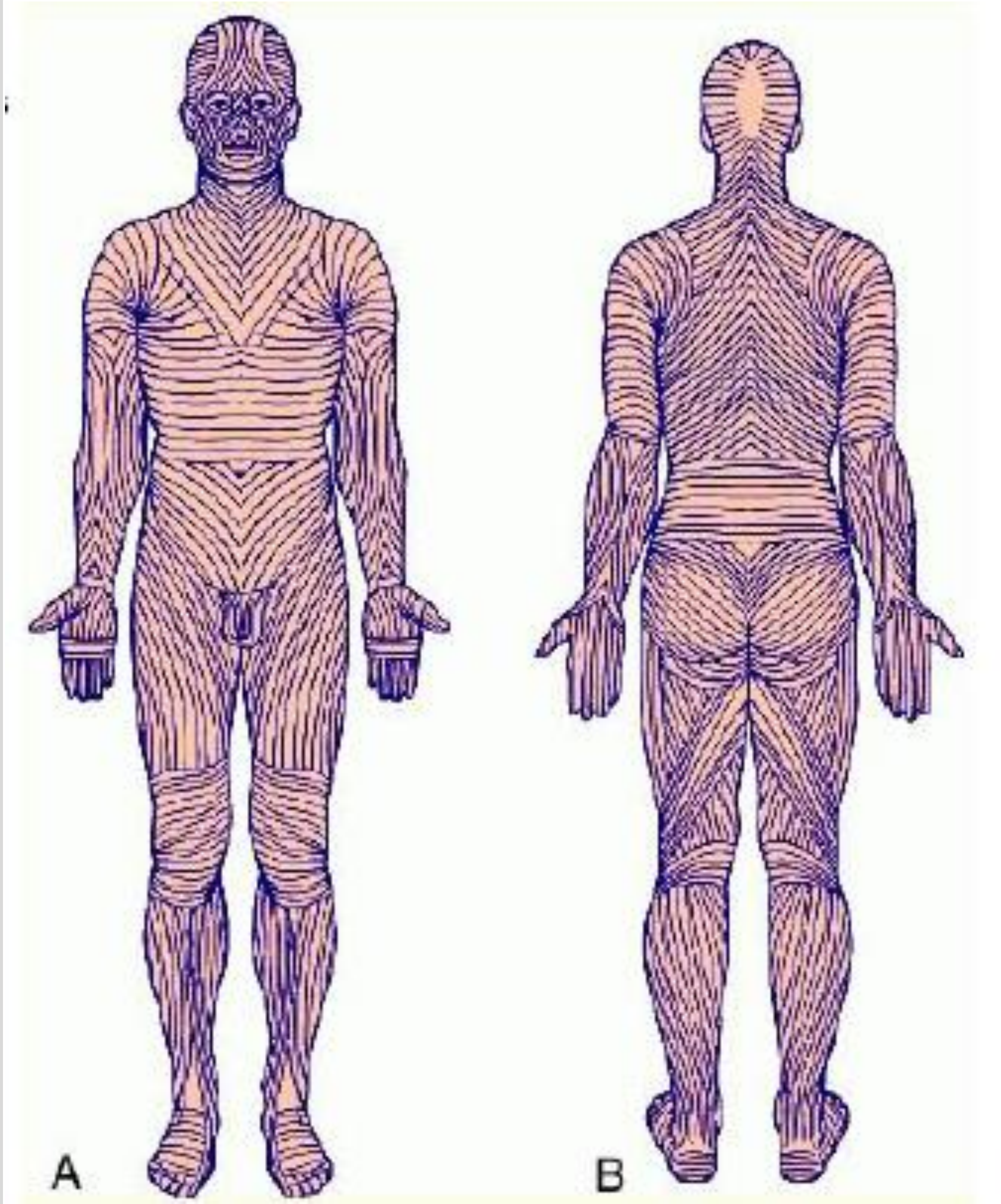
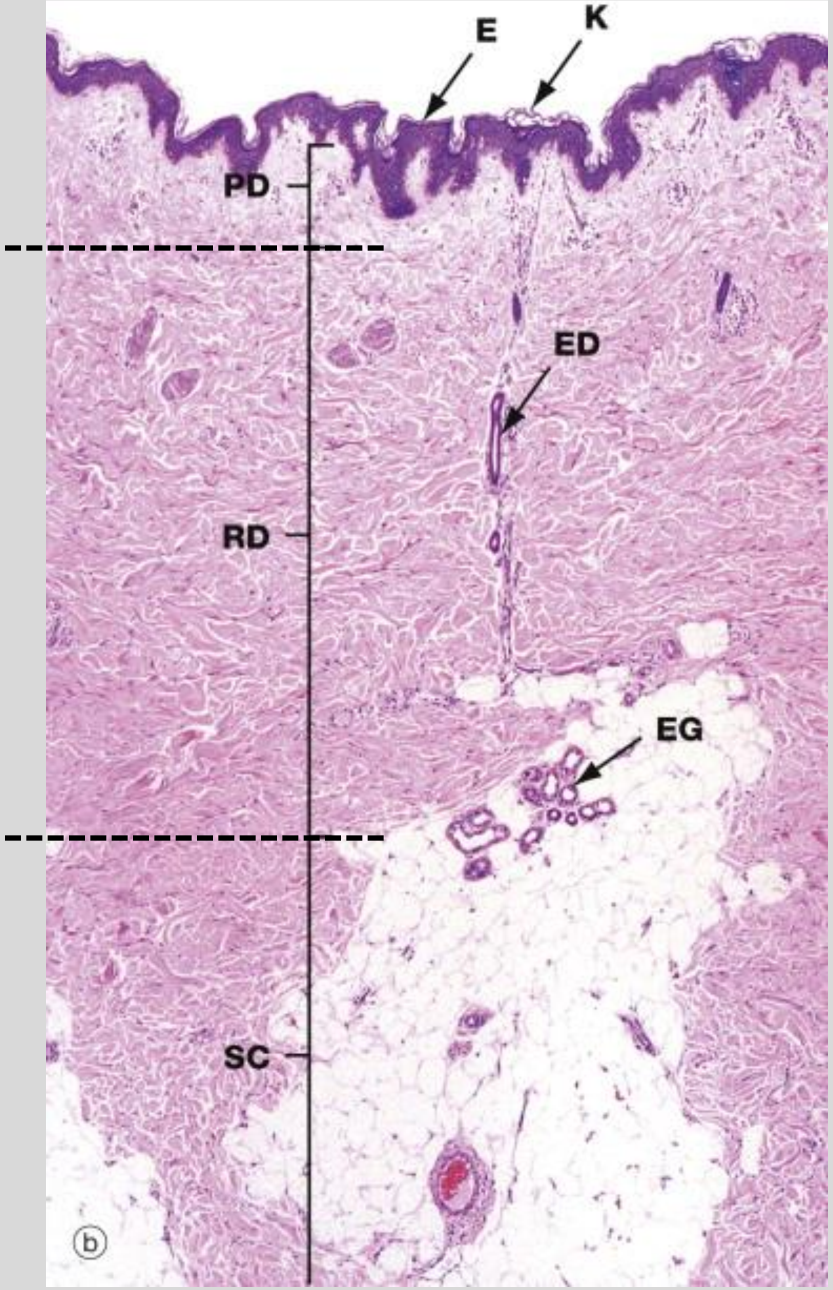


The collagen and elastic fibers are consistent orientation and form regular lines of tension.

Provide integrity

If skin incisions are made along these lines, there is less scarring. Cut across these lines and you get more scarring.

Hypodermis



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Also called subcutaneous fascia

Contains adipose lobules along with some skin appendages (hair follicles), sensory neurons, and blood vessels (highly vascular).

Important for energy storage

Important site for delivering injections

Source: “Skin” in Wheater’s Functional Histology, 6th Ed. (2014)

Practice Question #1

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Organize the following epidermis layers from most superficial to least superficial:

- Stratum Basale
- Stratum Corneum
- Stratum Spinosum
- Stratum Grangulosum

Practice Question #1

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Organize the following epidermis layers from most superficial to least superficial:

- Stratum Basale
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- Stratum Grangulosum

Most Superficial:	Stratum Corneum
	Stratum Granulosum
	Stratum Spinosum
Deepest:	Stratum Basale

Practice Question #2

A histologist is researching the molecule that is responsible for the desquamation of skin cells. What molecule is this, and what is it triggered by?

1) Kallikrein-related serine peptidase, lowered pH near the surface
2) Lymphoepithelial Kazal-type Inhibitor, lowered pH near the surface
3) Kallikrein-related serine peptidase, increased pH near the surface
4) Lymphoepithelial Kazal-type Inhibitor, increased pH near the surface

Practice Question #2

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Summary

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- Skin is a multi-function organ
- Epidermis is composed of continuously renewing keratinocytes expressed as different layers.
- Epidermis includes immune cells and melanocytes. Also serves as a water barrier.
- Dermis is connective tissue, vascularized, and houses epidermal appendages.
- Hypodermis is composed of adipose.

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Feedback is appreciated!

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R